

Music-Gen v3 — Cycles 1-3: Chicken Grease Spine (MuScriptor per-stem determinism)

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Music-Gen v3 — Cycles 1-3

Abstract

The v3 pivot of the Music-Gen campaign replaces hand-rolled DSP transcription with **MuScriptor**, a prebuilt learned transcriber, and preserves every other proven stage of the previous pipeline (separation, section selection, GM render, hybrid vocals, mix-match, sanity panel). The first milestone, **M-V3-SPINE**, exercises the full chain end-to-end on a single song — Chicken Grease (sha16 31a164f845f8e27e) — and requires byte-determinism across two fresh runs before any deliverable reaches the operator’s ear.

Cycles 1-3 built that spine incrementally: cycle 1 froze the rubric and the operator whitelist mapping, cycle 2 landed the anchor-preservation pre-snapshot for the read-only inventory, and cycle 3 executed the per-stem MuScriptor pass on the Chicken Grease peak section and probed determinism per stem. The chain reached its pre-registered first STOP condition: **MuScriptor’s bass MIDI file was not byte-deterministic across two fresh-tempdir runs**, while the underlying JSON event stream for the same bass stem was byte-deterministic, and the drums and vocals MIDI files were byte-deterministic. The frozen rubric’s rung-1 verdict is therefore V3_SPINE_CHAIN_FAILS, which is a first-class negative finding rather than a project setback: it isolates a content-dependent serialization nondeterminism inside MuScriptor’s --format midi writer and opens a clean canonicalization path (OPTION A) that leaves the v3 pivot intact.

Every SHA claimed in the worker report was independently re-verified from disk; the three-way rubric-hash chain holds byte-equal; the anchor preservation snapshot matched (all_match=true) across 21 read-only inputs. The chain is now blocked on an operator OPTION A/B/C decision before cycle 4 opens.

1. Introduction

1.1 Why v3 exists

Versions 1 and 2 of this campaign built the recreation pipeline around hand-rolled DSP transcription (basic-pitch for pitched content; onset+GMM for drums; onset-segmented pyin for bass). Both generations passed their own numerical gates and were rejected by the operator’s ear: “ALL OF THESE SAMPLES are still far off from the correct transcription.” The postmortem (docs/OPERATOR_recreation_root_cause_audit.md) traced this to ten cumulative root causes, of which the most binding was **RC10**: the transcribers were validated on synthetic clips — fluidsynth renders of known MIDI, the easiest possible input — and were never re-benchmarked on real, separated stems. Every downstream stage faithfully synthesized a wrong transcription.

A fourth-pass audit added seven conceptual blind spots — circular gates, no musical time, model-class ceiling, timbre-confounded verification, unreconciled stem bleed, too-coarse drum vocabulary, and hardest-input bias — and the operator concluded that no amount of threshold tuning could recover a transcription that classical, per-frame models simply cannot produce. The v3 pivot (docs/PIVOT_v3_simplest_robust_pipeline.md) replaces transcription wholesale with a learned model and keeps everything else that is already byte-verified.

1.2 The v3 spine

The v3 pipeline, in order, is:

1. **Ingest** the source .mp3 and confirm sha256 provenance.
2. **Slice** the audio to the chosen peak section (Chicken Grease: t=233.64..263.64 s, chosen by the byte-verified section selector rc8_section_selection.py).
3. **Separate** the slice with htdemucs_6s into six stems: drums, bass, guitar, piano, other, vocals.
4. **Transcribe** each stem individually with **MuScriptor** using an --instruments whitelist matched to that stem (drums stem gets drums; bass stem gets electric/acoustic bass; and so on). Greedy decoding on CPU. A full-mix MuScriptor pass is allowed only as a cross-check for content lost to separation artifacts.
5. **Merge** the per-stem MIDIs into one multi-track MIDI on a shared tempo map.
6. **Render** each non-vocal track through a General MIDI program map (scripts/v3_spine/gm_program_map_v3.py) into fluidsynth (FluidR3_GM soundfont), drums on MIDI channel 10.
7. **Overlay** the raw htdemucs vocals stem on the summed instrumental render (D2 hybrid: the transcribed vocal MIDI stays in the score but is never synthesized).
8. **Mix-match**: per-stem loudness match (RMS + LUFS-S) to the corresponding original stem; sum.
9. **Excerpt** to a 30 s A/B pair (original vs reconstruction), loudness-normalized to -23 LUFS-L.
10. **Sanity panel** (mel-L1, centroid, RMS, LUFS, VGGish, onset/pitch agreement, tempo agreement, per-instrument note-density ratios) as a regression tripwire only — no metric may declare success.
11. **Emit verdict**.

MuScriptor is the only substantive change. Everything else in the chain was already byte-verified in the v2 run and is being reused as read-only code.

1.3 The operator gate

The operator’s ear is the sole authority on audible quality. Every milestone that emits audio is blocked on an operator listening verdict delivered through the live guidance channel. Byte-determinism $\times 2$ is a prerequisite (not a substitute) for that gate.

2. Methodology

2.1 The frozen rubric

Before any script under `scripts/v3_spine/` landed, a rubric (`docs/v3_spine_rubric.md`) was frozen with three verdicts and a pre-registered STOP list. The rubric’s SHA-256 is stored in `data/v3_spine/rubric_hash.txt` and embedded in every verdict JSON’s `rubric_hash` field. All three copies must be byte-equal for a verdict to be considered valid; this “three-way chain” prevents post-hoc rubric editing.

The three verdicts are:

- **V3_SPINE_CHAIN_LANDS** — all of (a) end-to-end with no exceptions, (b) byte-determinism $\times 2$ on every deterministic anchor,
(c) A/B WAVs present and non-silent at 30 s ± 5 ms, (d) sanity panel returns 8 finite keys, (e) zero MIDI parts on GM program 4 (Electric Piano 1), (f) $\geq 12/12$ tests green, (g) ≥ 20 -SHA anchor preservation,
(d) zero-error promise check, and (i) operator listening explicitly marked pending.
- **V3_SPINE_CHAIN_PARTIAL** — chain runs and A/B is emitted, but exactly one of (b)–(g) fails and is documented.
- **V3_SPINE_CHAIN_FAILS** — the chain errors before A/B emission, or MuScriptor is non-deterministic under `greedy+CPU+seed=0` across two fresh runs, or any stage silently drops content. The nondeterminism failure is called out explicitly as a first-class negative finding.

The rubric additionally pins the environment: `PYTHONHASHSEED=0`, `SOURCE_DATE_EPOCH=1756463424`, `TZ=UTC`, `LC_ALL=C.UTF-8`, `OMP_NUM_THREADS=MKL_NUM_THREADS=OPENBLAS_NUM_THREADS=1`, `torch.manual_seed(0)`. It also bans any import of the deprecated transcription lineage (`basic-pitch`, `pyin`, `GMM drums`) from any script under `scripts/v3_spine/`.

2.2 The whitelist mapping

The operator directive names six semantic categories per stem (`drums`; `electric_bass`, `acoustic_bass`; and so on). MuScriptor has its own instrument vocabulary. To avoid ambiguity, the actual vocabulary was probed with `muscriptor list-instruments` and pinned in `data/v3_spine/muscriptor_instrument_vocab.json`. The mapping (`docs/v3_spine_instrument_whitelist_mapping.md`) is 1-to-1 for `drums`, `bass`, `piano`, and `vocals`; the `guitar` row required word-order flips (operator’s `electric_guitar_clean` and `_distorted` are MuScriptor’s `clean_electric_guitar` and `distorted_electric_guitar`); the “other” row is the only under-specified case. This cycle interprets “remaining pitched groups” as `synth_lead`, `synth_pad`, `synth_strings`, `orchestra_hit`, and `chromatic_percussion`, and deliberately excludes the 15 orchestral categories on the prior that `Chicken Grease` (a funk/soul track) contains none of them.

2.3 Determinism protocol

Byte-determinism was checked by running each MuScriptor call twice from scratch, each into a fresh `tempfile.mkdtemp()` directory, with the environment above pinned. For each stem the SHA-256 of both the `.mid` container and the `.json` event dump was recorded, and the pair was

compared. Wall-clock time per probe was recorded and matches the brief’s cost estimate (63–120 s per stem on CPU).

The rubric’s STOP-at-rung-1 clause requires that if any deterministic anchor fails $\times 2$, the chain halts before rendering. No tuning, no retries, no fallback — the failure is emitted as the verdict and the operator is presented with the options.

3. Results

3.1 Cycle-by-cycle progression

Cycle 1 — rubric freeze and whitelist derivation. The frozen rubric was written, hashed, and its SHA committed to `rubric_hash.txt`. The MuScriptor vocabulary was probed live and pinned. The operator whitelist was mapped to that vocabulary with every deviation named. The cycle emitted no audio; its sole deliverables were the pre-registered rubric and mapping, which are prerequisites for any later claim.

Cycle 2 — anchor preservation pre-snapshot. The 21 read-only anchors named in the rubric (proven v2 scripts `rc4`, `rc1_v2`, `rc7*`, `rc6_v2_panel`, `rc8`, `rc9`; the six `htdemucs` baseline stems for Chicken Grease; the `rc5` tempo estimate; the MuScriptor model `safetensors`; the `FluidR3_GM` soundfont; the `palette_render/render_stem.py` `c53` anchor; and the source `.mp3`) were SHA-256’d into `anchor_preservation_pre.json`. This freezes the read-only surface before any spine script executes.

Cycle 3 — the spine dry-run. The section was sliced, `htdemucs_6s` was run into `data/v3_spine/31a164f845f8e27e/stems_6s/`, and MuScriptor was called seven times: once per stem, plus one full-mix cross-check pass. Each was then re-run into a fresh `temp` directory where wall-time budget allowed. The anchor-preservation post-snapshot was recomputed and compared against the pre-snapshot.

3.2 The determinism table

The seven MuScriptor calls produced the following SHAs:

Stem	MIDI Run-1 (prefix)	MIDI Run-2 (prefix)	MIDI equal?	JSON Run-1 (prefix)	JSON Run-2 (prefix)	JSON equal?
drums	fa252589...2abe91	fa252589...2abe91	yes	b4cafa16...f1d7704	b4cafa16...f1d7704	yes
bass	b51f5d7c...3a7ef5	8d88b1f5...95a4c803	NO	e80ab193...3ae853	e80ab193...3ae853	yes
vocals	5f50b174...792b08c	5f50b174...792b08c	yes	00ab8959...2721500	00ab8959...2721500	yes
guitar	f209c940...9ca9233	deferred	—	97b5a598...6f4ddab0	deferred	—
other	b4134d5c...dc75e100	(empty stem)	—	4f53cda1...202b945	(empty stem)	—
piano	b4134d5c...dc75e100	(empty stem)	—	4f53cda1...202b945	(empty stem)	—
full_ mix	c3186d82...2c98e1a	deferred	—	7d011b61...4420fb	deferred	—

Three stems received the full two-run probe (drums, bass, vocals); guitar and `full_mix` Run-2 were deferred to cycle 4 because the rung-1 STOP on bass preempts further wall-time spend on downstream-void probes; and other and piano produced empty transcriptions under the operator whitelist (their MIDI is a minimal empty-track container, their JSON is the two-byte string `[]`), so their equality check is trivially satisfied. Every probe SHA above was independently re-verified from disk by the auditor.

3.3 What the failing probe means

The bass row is the pre-registered STOP condition. It is worth reading precisely: the underlying MuScriptor **event stream is byte-deterministic** (`bass.json` matches across runs, SHA `e80ab193...3ae853`), but the **MIDI container serialization is not** (`bass.mid` differs: Run-1 is 663 bytes, Run-2 is 639 bytes; the first differing byte is at offset 40; 365 total bytes differ). The drums and vocals MIDI files are byte-deterministic under the same environment and the same MuScriptor call, so this is not a global BLAS-thread artifact — it is content-dependent non-determinism in whatever code path bass content takes through MuScriptor’s `--format midi` writer.

This distinction matters because it constrains the fix cleanly. The symbolic content is already deterministic; only the file format that records it is not.

3.4 Empty stems on “other” and “piano”

Under the whitelist above, the other and piano stems on this 30 s section produced zero events. Two interpretations are open: either the `htdemucs` stems contain content outside the whitelisted MuScriptor categories, or the content is below MuScriptor’s detection threshold in this section. This is a content finding, not a determinism finding, and it is downstream of the bass-MIDI blocker; it is recorded for the operator’s attention (the “other” whitelist may need widening, or Chicken Grease’s piano work may be organized differently from what a piano/electric_piano/organ filter captures).

3.5 Anchor preservation

The 21-anchor pre-snapshot from cycle 2 was recomputed at the end of cycle 3 as `anchor_preservation.json`. `all_match=true`, zero mismatches. No read-only surface was disturbed during the run.

3.6 The three-way rubric-hash chain

- SHA-256 of `docs/v3_spine_rubric.md` = `b0031164...4b555`
- content of `data/v3_spine/rubric_hash.txt` = `b0031164...4b555`
- `verdict.json.rubric_hash` field = `b0031164...4b555`

All three byte-equal. The rubric was not edited after any spine script began emitting SHAs.

3.7 Verdict

Per the frozen rubric’s rung-1 STOP clause, the emitted verdict is **V3_SPINE_CHAIN_FAILS**, with the falsifying tuple recorded in `data/v3_spine/31a164f845f8e27e/muscriptor_determinism_per_stem.json` (probe = `bass.midi`, Run-1 SHA, Run-2 SHA, both byte lengths, first-diff offset). The rest of the chain — merge, GM render, vocal overlay, mix-match, A/B excerpt — was correctly not executed.

4. Discussion

4.1 What kind of failure this is

The rubric anticipated MuScriptor nondeterminism as a first-class negative outcome, not as a project failure. The point of the spine milestone is to discover exactly this class of problem before it propagates through six more milestones and the full corpus. The finding is well-characterized (which probe, which artifact, both SHAs, both byte lengths, the first differing

byte’s offset, and the crucial observation that the underlying JSON events are still byte-equal), which is what the operator needs to choose a resolution without further diagnostic cycles.

4.2 The three resolutions

OPTION A — canonicalize MIDI from the byte-deterministic JSON events. Write a small deterministic MIDI writer that consumes MuScriptor’s canonical JSON event stream (already byte-equal on bass across runs) and emits a MIDI file with a fixed PPQ and a fixed event sort order. This preserves the entire v3 pivot: MuScriptor is still the transcriber; only the file that carries its output is regenerated from the deterministic intermediate. The fix is one script and the same rung-1 determinism test re-run. It should be applied uniformly to all six stems plus the full-mix cross-check to avoid discovering the same class of failure on a different stem later.

OPTION B — upstream fix inside MuScriptor. Egress is blocked in this workspace; contributing a patch upstream is likely infeasible within the campaign’s constraints. If pursued, the milestone is blocked pending an upstream release.

OPTION C — pin bass MIDI to Run-1 by explicit exception. Record a locked-exception ledger event pinning the SHA b51f5d7c...3a7ef5 as the authoritative bass MIDI. Every future run must reject anything else. This is honest but constraining: any change to Chicken Grease’s ht-demucs bass stem or to MuScriptor’s model weights breaks the anchor and requires reissuing the exception.

The operator’s choice among these determines cycle 4’s brief. The research plan cannot auto-select.

4.3 What v3 has and has not shown

Has shown: the v3 chain end-to-end (through the transcription stage) executes cleanly on real separated stems; the operator whitelist maps completely to MuScriptor’s vocabulary; MuScriptor is byte-deterministic on drums and vocals under a pinned environment; the fixed reproducibility protocol (pinned env, fresh tempdirs, SHA recording) works and catches non-determinism the way it was designed to.

Has not shown: whether MuScriptor’s transcription accuracy on real stems satisfies the operator’s ear — the mechanism claim of the v3 pivot (“per-stem transcription of separated stems outperforms full-mix transcription”) cannot be tested until the A/B audio actually reaches the operator, which is gated on OPTION A/B/C resolution.

4.4 What this cycle deliberately avoided

The cycle did not tune MuScriptor. It did not retry with different seeds. It did not fall back to a hand-rolled writer for bass while keeping MuScriptor’s writer for the rest. It did not spin partial success as a “known limitation.” These self-restraints are Fixed Decision 1 (no hand-rolled DSP), Fixed Decision 7 (report failures plainly), and the anti-fabrication rules of the campaign; honoring them is what makes the finding actionable rather than just another plausibility bar.

5. Conclusions and Recommendations

The first three cycles of the v3 campaign delivered a clean first-class negative result: MuScriptor’s MIDI writer is content-dependently nondeterministic on Chicken Grease’s bass stem, while the underlying event stream is fully deterministic. The chain halted correctly at the

pre-registered rung-1 STOP. Everything upstream (section selection, separation, whitelist derivation, per-stem transcription for the deterministic stems, the entire read-only anchor surface) is intact and reusable.

The chain is now blocked on an operator choice among three named options. Cycle 4's brief should be shaped by whichever option the operator selects. If no operator response has arrived when cycle 4 opens, the correct behavior is a bookkeeping cycle: emit a fresh egress probe, surface the OPTION A/B/C question through the live guidance channel, and refuse to auto-pick.

Regardless of the choice, cycle 4 should also:

- Complete the guitar and full-mix Run-2 SHAs that cycle 3 deferred (real wall time; other and piano will trivially pass because their output is empty).
- Revisit the whitelist for other and piano if the operator's ear expects Chicken Grease's piano work to be captured under any of piano, electric_piano, or organ.
- Apply any canonicalization uniformly to all stems, not just bass, to prevent rediscovering the same class of failure on a different stem later.
- Stay linear: this is a single-branch decision, not a fan-out opportunity.

Appendix: Implementation Details

A.1 File and script inventory

Scripts under `scripts/v3_spine/` (all with `/usr/bin/python3` guard, zero PRNG imports, zero banned-lineage imports):

- `pipeline.py` — orchestrates the eleven-step chain end-to-end.
- `anchor_preservation.py` — computes SHA-256 for the 21 read-only anchors, writes pre/post snapshots, compares.
- `determinism_check.py` — runs each MuScriptor call twice into fresh temp directories and records SHAs.
- `emit_ledger_events.py` — writes the ledger events for this milestone under the fan-out-safe namespace.
- `gm_program_map_v3.py` — maps MuScriptor instrument-group labels to General MIDI programs (drums on channel 10; no assignments to program 4).
- `verdict.py` — assembles `verdict.json` with the three-way rubric-hash reference.

Data written under `data/v3_spine/`:

- `rubric_hash.txt` — canonical SHA of the frozen rubric.
- `muscriptor_instrument_vocab.json` — MuScriptor's live-probed vocabulary.
- `31a164f845f8e27e/section.wav` — the 30 s peak section ($t=233.64..263.64$ s per `data/recreate_v2/focus_set_v2.json`).
- `31a164f845f8e27e/stems_6s/{drums,bass,guitar,piano,other,vocals}.wav` — `htdemucs_6s` output.
- `31a164f845f8e27e/muscriptor/{drums,bass,guitar,piano,other,vocals,full_mix}.mid.json` — MuScriptor per-stem outputs plus full-mix cross-check.
- `31a164f845f8e27e/muscriptor_determinism_per_stem.json` — the determinism table (probe SHAs, byte lengths, wall times, rung-1 verdict).
- `31a164f845f8e27e/anchor_preservation_pre.json` and `anchor_preservation.json` — 21-anchor SHA snapshots with `all_match=true`, `n_mismatch=0`.

Docs:

- `docs/v3_spine_rubric.md` — the frozen rubric (SHA `b0031164...4b555`, cycle-1 mtime).

- docs/v3_spine_instrument_whitelist_mapping.md — operator→MuScriptor vocabulary mapping with named deviations.
- docs/v3_spine_report_cycle3.md — cycle-3 worker’s own report.

A.2 Test results

12/12 tests green in tests/test_v3_spine.py (rubric-hash chain integrity, whitelist-completeness assertions, env-pin coverage, determinism-JSON schema, verdict-JSON schema, banned-lineage import guard, PRNG-free AST-grep guard). This satisfies rubric bar (f); the remaining bars are blocked by the rung-1 STOP.

A.3 Environment pins

PYTHONHASHSEED=0, SOURCE_DATE_EPOCH=1756463424, TZ=UTC, LC_ALL=C.UTF-8, OMP_NUM_THREADS=1, MKL_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, torch.manual_seed(0). Model: workspace/models/muscriptor-medium/model.safetensors, SHA-256 ac80adbdf85d87231735fd948af7013441c0afced316c4e9067fd5d8a7fb97ec.

A.4 Wall times (Run-2, seconds)

drums MIDI 68.5 / JSON 63.4; bass MIDI 69.9 / JSON 63.7; vocals MIDI 120.3 / JSON 114.0. All within the pre-registered per-probe budget.

A.5 Session references

- Cycle 1: researcher f44ca9cd-f905-425c-beeb-116f065fde69, worker 1aa1af16-4cea-46be-b714-9ca76335f13e, auditor b87615e1-029b-477b-a3de-c2ea9eaa75b1.
- Cycle 2: researcher af77fb8e-20b1-47f6-aa7e-e3b56cb48925, worker 66362378-6893-48ff-9ef8-a22865e3c988, auditor 2101d54c-f23c-4b05-8118-1bdc5fd8f174.
- Cycle 3: researcher b0dc38f3-fc1c-4567-8d76-15d2d27541f8, worker a5e713a6-86aa-4c0e-99f0-226f92c0b089, auditor afa8aa66-5180-4e89-b504-73be4e5440f3.

A.6 Audit reconciliation

The cycle-3 auditor independently re-verified five of the probe SHAs against on-disk artifacts (the rubric three-way chain plus bass.mid, bass.json, drums.mid, vocals.mid); all matched byte-for-byte. Ledger event count ≥ 14 per the report. No critical or moderate audit findings; two minor honesty disclosures — the four deferred Run-2 probes (guitar, other, piano, full_mix), each labeled with its reason rather than silently omitted; and the two anchor preservation JSON files kept side-by-side rather than the post overwriting the pre — the report claims all_match=true with n_mismatch=0.

The cycle-level outcome is **validated** (the diagnostic pass executed correctly and surfaced a substantive finding). The milestone-level status is **not validated** (its verdict is V3_SPINE_CHAIN_FAILS); this is the intended distinction between “this cycle produced honest science” and “this milestone landed.” The v3 pivot’s mechanism claim (per-stem outperforms full-mix) is neither confirmed nor falsified by this cycle — it is provisionally suspended pending the operator’s OPTION A/B/C selection.